Rakt Setu - An Integrated Mobile Platform for Expediting Blood Donation and Request Management

Ayush Srivastava
Department of Computer Science and Engineering(Artificial Intelligence and Machine Learning)
Krishna Institute of Engineering and Technology (KIET)
Ghaziabad, Delhi-NCR, Uttar Pradesh, India
srivastava.ayush6220@gmail.com

Shaurya Jain
Department of Computer Science and Engineering(Artificial Intelligence and Machine Learning)
Krishna Institute of Engineering and Technology (KIET)
Ghaziabad, Delhi-NCR, Uttar Pradesh, India
shauryajain236@gmail.com

Prapti Sharma
Department of Computer Science and Engineering(Artificial Intelligence and Machine Learning)
Krishna Institute of Engineering and Technology (KIET)
Ghaziabad, Delhi-NCR, Uttar Pradesh, India
praptisharma282003@gmail.com

Shriyansh Srivastava
Department of Computer Science and Engineering(Artificial Intelligence and Machine Learning)
Krishna Institute of Engineering and Technology (KIET)
Ghaziabad, Delhi-NCR, Uttar Pradesh, India
shriyanshsrivastava516@gmail.com

Mr. Thammali Gangadhar
Department of Computer Science and Engineering(Artificial Intelligence and Machine Learning)
Krishna Institute of Engineering and Technology (KIET)
Ghaziabad, Delhi-NCR, Uttar Pradesh, India
thammali.gangadhar@kiet.edu

***Abstract*— The critical shortage of timely blood supply in emergency medical situations is a significant global healthcare challenge. This challenge becomes more serious due to poor communication, logistical difficulties, and the absence of real-time coordination between donors and seekers. The primary objective of this research is to present a high-performance, integrated web application – Rakt Setu, that automates the end-to-end blood request management process through real-time geospatial coordination. Rakt Setu's architecture uses MERN (MongoDB, Express.js, React, Node.js) stack to guarantee a full-JavaScript, scalable, and maintainable codebase. The website has a dual-role architecture in which seekers create urgent, location-based requests to find suitable donors or blood banks. The smart geospatial matching tool, built into the platform and reliant on the MongoDB in-built geospatial query tool to alert nearby compatible donors immediately, is one of the innovations. While the Node.js backend is used to handle real-time APIs, the frontend is a single-page application (SPA) that is built on React. The entire application is bundled together with Docker, which provides a consistent environment of development and deployment. When one accepts a request, a secure, two-way, exchange of contact information is made, which guarantees direct communication. The application also connects to the Google Maps API to allow travelling and cooperates with certified hospitals to offer safe and authorized spots of donation. The long-term goal of this end-to-end workflow is to significantly lower the response time in emergencies and enhance donor participation as well as overall effectiveness of blood supply chain between request origination and assisted donation.**

***Keywords*— Blood donation, MERN stack, Real-time Notification System, Geospatial Query, Docker containerization, Healthcare Logistics.**

I. INTRODUCTION

BLOOD is an indispensable, life-saving resource in modern medicine, crucial for treating trauma victims, surgical patients, and individuals with chronic conditions. However, many regions face chronic shortages due to seasonal donor scarcity, inefficient inventory management, and a lack of coordinated communication. Addressing this blood deficit is essential to help avoid delays in critical blood transfusions for people in need. The proliferation of smartphones presents a transformative opportunity to address these inefficiencies through dynamic, location-aware, and real-time coordination.

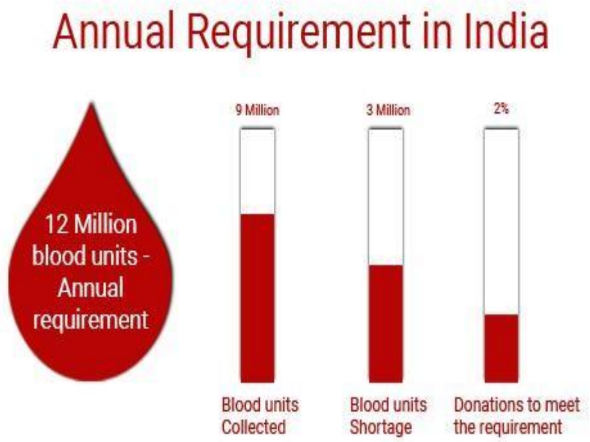

Fig. 1. Blood Shortage in India [1]

While various digital solutions have been introduced, a critical gap analysis of earlier studies reveals significant operational weaknesses. Early platforms such as "Friends2Support" [3] established large databases which remained fundamentally passive, requiring seekers to manually contact individual donors which is a time-consuming process and often fails during the "Golden Hour" of medical emergencies. Institutional portals like "E-Rakt Kosh" [5] have improved inter-bank coordination but do not enable direct donor-seeker matching for urgent needs. Existing mHealth apps, such as the Red Cross Blood Donor app [4], primarily function as one-way notification systems rather than interactive ecosystems. Furthermore, reliance on unregulated social media crowdsourcing does not have a structured response mechanism and has privacy risks. This research proposes a holistic, closed-loop mobile ecosystem built on the MERN stack to bridge these identified gaps.

The primary contribution of this work is an integrated platform – "Rakt Setu", which is built on the MERN stack that distinguishes itself through:

- A dynamic, proximity-based matching algorithm based on MongoDB's geospatial indexing.
- A secure, bidirectional communication protocol driven by a backend built with Node.js and Express.js.
- A seamless user experience is provided by a responsive React-based frontend.
- A containerized and globally deployed architecture using Docker, ensuring scalability and reliability.

This paper details the MERN-based architecture, functionality, and the Docker deployment of Rakt Setu.

II. Literature Review

Three primary areas of research have been conducted on the use of information technology to optimize blood supply chains: integrated mobile health (mHealth) apps, inventory and supply chain optimization, and donor management systems.

A. Donor Management and Recruitment Systems

A substantial amount of research focuses on methods for managing, recruiting, and retaining donors. In North India, Sachdev S, Mishra SK, Marwaha N, and Avasthi [6] carried out a study to determine college-bound students' attitudes about volunteer blood donation, including information gaps and inconvenient times. Early alternatives included digital platforms such as "Friends2Support" [3], which established online databases of voluntary donors. Although they are effective at creating vast databases, their main drawback is that they are passive; in an emergency, seekers must personally contact several donors from a list, which is a time-consuming and inefficient operation. They don't have automatic, intelligent matching or real-time availability checks.

B. Inventory and Supply Chain Management

There are various research works which focus on the logistics of blood inventory management after collection. For example, "E-Rakt Kosh" portal [5] of the Indian Government provides a centralized view of blood inventory across numerous banks, significantly improving inter-bank coordination. Osorio et al. [7] modelled and optimized the logistics of the blood supply chain using discrete-event simulation. It focused on reducing waste and improving distribution effectiveness. Although such platforms are important, they operate at the institutional level and do not help in quickly locating a donor for a specific, urgent need that is not covered by the current inventory.

C. Integrated mHealth Applications

The rise of smartphones led to the creation of mobile applications and websites for blood donation. The Red Cross has a site called Blood Donor [4] that assists the user in locating blood drives and making appointments. It functions largely as a transmission system in terms of blood banks to donors. With a narrower target, the BloodLink Navigator System [8] was developed to enhance blood-request system and streamline the process of blood donation. This system offers an effective platform where the seekers, donors, and administrators can handle blood-related events effectively.

The challenges of adopting these systems have been researched. It has been discovered that a mistrust and the absence of perceived reliability in India makes it difficult to implement blood donation applications. They indicated that collaboration with reputable organizations, such as famous hospitals, could be used to deal with such problems. Also, some of the suggested systems such as Blood donor searching android application with GPS tracking by Sangeetha M., Dhivya A., Ramya R., and Sharmila N., [10] is intended to provide the ability to view the details of the nearby donors and easily connect with the donors in case of an emergency. This platform does not however publish the information of the donor to the seeker within 90 days hence impedes two way communication.

D. Social Media and Crowdsourcing

Social media platforms like Facebook and Twitter have become popular ways to request blood. They help a seeker to send a blood request to lakhs of people in one go. Facebook's "Blood Donations" feature lets users sign up as donors and get alerts about nearby needs. This method benefits from its large network. However, it is an unregulated, unreliable process that involves various privacy concerns and relies on the popularity of a post. Thus, social media platforms do not ensure a structured and guaranteed response towards a blood request.

E. Identified Research Gap

Previous research works and the existing platforms have made important contributions, yet they do not provide real-time donor matching and secure communication. Our proposed platform – Rakt Setu aims to fill this gap by creating a managed ecosystem that connects an urgent blood request with a successful donation. It works to connect a donor and a seeker directly without involving any third-party or middlemen.

Feature	Passive Directories (e.g., Friends2Support [3])	Institutional Portals (e.g., E-Rakt Kosh [5])	Social Media / Crowdsourcing	Proposed System (Rakt Setu)
Operational Model	Manual search from list	Institutional inventory	Unregulated broadcasting	Real-time seeker-donor matching
Matching Engine	None (Static database)	Centralized stock view	Popularity/Algorithm-based	Automated Geospatial (2dsphere)
Communication	One-way/Seeker initiated	Inter-bank coordination	Unregulated/Private DMs.	Secure Bidirectional Exchange
Privacy Layer	High exposure of data.	Secure (Institutional)	Significant privacy concerns	Contact exchange only on acceptance
Logistics Support	None	General location mapping	None	Integrated Google Maps Navigation

Table. 1. Comparative Evaluation

III. Research Methodology

This research follows a structured Software Development Life Cycle (SDLC) Prototyping approach. Unlike traditional methods, this allows for the creation of a working model which can be refined until it meets all the necessary requirements of the web app.

A. The Four Research Stages

1. Requirement Analysis – A comprehensive analysis was performed to identify the reasons for failure of existing platforms and the essential requirements for an effective blood donation management platform. We reviewed previous studies and found that "communication latency" - the delay after which the donor receives a message, is the biggest hurdle in blood supply chains.

2. System Design & Data Modelling – The second phase of the research process is the drawing of blueprints of the app. A dual-role system has been implemented in which users are made different in case they are a Donor or Seeker. On MongoDB, we designed a certain data schema using geospatial indexing. This enables the database to consider the position of a user as a point on a map and not a piece of text. This enables milliseconds distance-based searching.

3. Implementation & Integration - This is the construction stage with MERN stack. We have also incorporated Google Maps API to guide the donors to donation points. We relied on the Firebase (FCM) to dispatch "Push Notifications" that appear on the phone of a donor immediately.

4. Empirical Testing – It is the final stage and here we measured the speed of the system. We attempted the time-consumption of the query to locate a donor with a radius of 15-20km (Query response time). We were also able to measure the time interval of clicking "Request" on a Seeker to the phone of the Donor receiving a message (Notification Latency).

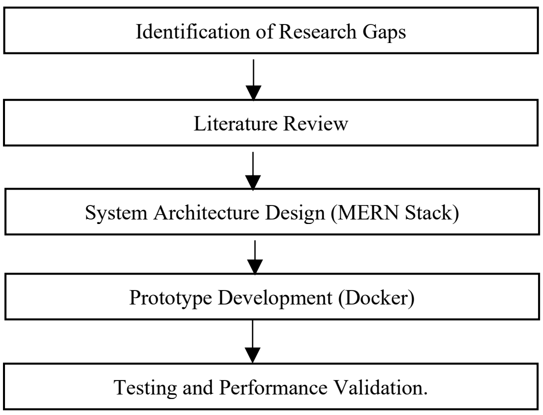

Fig. 2. Research Methodology Block Diagram

B. Model Parameters

Stage	Key Parameter	Purpose
Authentication	Auth Protocol (OTP)	To ensure user authenticity
Search Logic	2dsphere query	To find the nearby donors within a specified range
Notification	Multi cast push	To send one alert to many people at the same time
Donor Safety	90-day time interval	To ensure mandatory interval between two donations
Deployment	Docker containers	To ensure that the app runs on any system or server

Table. 2. Model Parameters

C. Procedural Workflow

Phase 1: User Registration and Data Collection

The first step is onboarding users on our platform, especially donors . Individuals can register themselves on the platform, selecting a specific role: Donor or Seeker. For building a comprehensive profile, the users provide essential details which include - personal information, contact information, their blood group, and the pin code (location data). Apart from individuals, blood banks can also register themselves on our platform by providing their contact details. The registration process is supported by OTP based user verification to build authenticity.

Phase 2: Request Initiation and Intelligent Matching

When a Seeker requires blood, they submit an urgent request through the platform. This request is immediately processed by the system's backend server, which identifies all registered donors as well as the blood banks who have the matching blood type, and are located within a predefined geographical radius of the request. The system then automatically sends a real-time push notification to these matched donors simultaneously. The Seeker also gets a list of Blood Banks along with their contact details, which have the requested blood type available in their stock. Once a seeker submits a request, he cannot make another request until and unless the previous request has been fulfilled. This assists in reducing the number of blood requests being made by one person at the same time.

Phase 3: Donor Response and Secure Communication

The donors that have their blood group matching that required are notified by the platform and the mobile devices and via the e-mails. The donors contacted can accept or reject the request depending on their location. In case the donor accepts the request, two-way sharing of contacts information (via direct SMS/ emails) takes place between the seeker and the donor. This move will not reveal any information about the donor or the recipient hence will not expose the information of the involved parties too early. The site also gives directions to the donor to the point where the seeker wants.

Phase 4: Donation Fulfilment and System Update

The last stage of the working process is real blood donation which is organized by hospitals/NGOs and updation of the database. Our platform provides the donor with the assistance of navigation using Google Maps and helps him reach the hospital, where the usual blood donation processes are conducted. After the donation process is done, the database of the platform is updated. The donor is then stipulated as not available to further donations temporarily over a given time period, in order to adhere to mandatory health intervals of donations. The donor dashboard has now shown the donation that has been made and the seeker is permitted to make new blood requests.

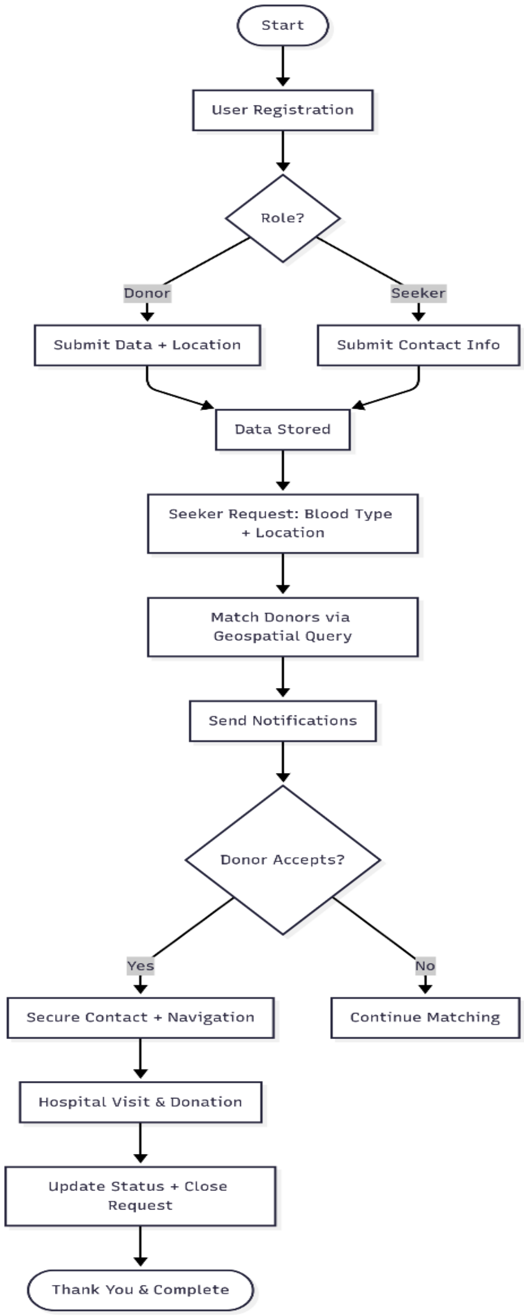

Fig. 3. Process Flowchart

IV. SYSTEM ARCHITECTURE AND METHODOLOGY

Rakt Setu has been built on the MERN stack architecture, which is a cohesive full-stack JavaScript framework. It is widely used in building dynamic web applications due to its great efficiency, scalability, maintainability, and performance.

A. User Roles and Data Models

Our system has two primary user roles with tailored data models:

1 - Donors: Donors are the users who are willing to donate blood voluntarily. Detailed information such as name, age, gender, contact info, address, and medical data (blood group, last donation date etc) is provided by the donors during registration. This data is stored securely and is essential for intelligent matching during urgent blood requests. (Table1.)

2 - Seekers: Seekers are those users who urgently need blood during emergencies. Seekers can register on our platform or use it as guests (for one time use) to generate blood requests. They too provide essential information such as the required blood type, the patient's location and their contact number, in order to submit a request.

Field	Description
donor_id	Unique donor ID
name	Full name
age	Age of donor
gender	Gender (M/F/Others)
blood_group	Blood group (A+/A-/B+/B-/O+/O-/AB+/AB-)
email	Email ID
contact_info	Phone number
address	Residential address
pincode	Pincode
password_hash	Encrypted passwords
fcmToken	Firebase token for notifications

Table. 3. Donor table

B. Technology Stack (MERN)

The system is implemented using MERN stack and the breakdown of technologies is discussed below:

1 - Frontend (React): The user interface of the web application is built using React. Multiple React libraries have been used to develop the frontend of the system. To illustrate, React router is an application routing the traffic between the various views (Registration page, Donor Dashboard, Seeker Dashboard, Blood Bank Page) of the webpage and Axios is employed in the smooth operation of HTTP communication with the backend API. The UI is responsive and has a native application-like behaviour on both the mobile and desktop browsers.

2 - Backend (Node.js + Express.js): Node.js and Express.js have been used to develop the backend logic of the web app. The backend layer is responsible for handling all HTTP requests from the frontend, implementing user authentication and authorization using JSON Web Tokens. It also executes the business logic, including the donor-matching algorithm and communicating with the database via Mongoose, an Object Data Modelling library for MongoDB.

3 - Database (MongoDB): MongoDB is a NoSQL document database. We use MongoDB for database implementation as it is known for its flexibility and native support for geospatial data. Here, the user locations are stored as GeoJSON objects, which allow for highly efficient proximity-based queries. A 2dsphere index is created on the location field to optimize these queries.

C. *Deployment and DevOps: Docker & Vercel*

We have used a modern DevOps approach to ensure consistency and scalability of the application:

Docker Containerization: The entire web-application is containerized using Docker. A Docker file defines the environment for the Node.js backend, and docker-compose.yml can be used to orchestrate the backend and MongoDB services together. This guarantees that the application runs identically across all environments—development, testing, and production—eliminating the "it works on my machine" problem.

Frontend Deployment: The React frontend is deployed on Vercel. Vercel provides:

- A Global CDN: The frontend assets are served from locations closest to the user worldwide, ensuring fast load times.
- Serverless Functions: The API routes can be implemented as Vercel Serverless Functions. However, currently we have the Express server maintained separately for WebSocket and real-time capabilities.
- Automatic CI/CD: Connecting the GitHub repository to Vercel triggers automatic deployments on every git push, streamlining the development workflow.

Backend Deployment: We can easily deploy the Node.js/Express backend API on platforms like, Render, Vercel or a cloud VM (e.g., AWS EC2) that supports running Docker containers. Environment variables are used to manage configuration settings, such as API keys securely across different deployment stages.

D. *Core Workflow*

The operational workflow is the core of the platform. All the processes follow a specific sequence of execution and are designed to ensure minimal latency. (Fig6.)

1 - Request Generation: The seeker submits a blood request through a structured form available in the Seeker dashboard. The frontend application then sends a request containing the JSON payload (blood type, location) to the /api/requests endpoint on the backend API server, initiating the emergency workflow to find the most suitable donors.

2 - Proximity Matching: The Express server receives the HTTP request and invokes the corresponding controller function. This function executes a geospatial query on the donor collection, to filter out available donors with a matching blood type available in the mentioned pin code or nearby areas which lie within acceptable radius, ensuring millisecond response times.

3 - Real-Time Notification: After successful execution of the geospatial query, the server retrieves the list of matching donors. A high-priority, multicast push notification is sent to all the devices associated with the donors' stored fcmTokens. The matched donors receive this notifications on the application as well as on their e-mails. This ensures immediate alerting without the need for the donor to have the application open.

4 - Donor Acceptance & Secure Contact Exchange: The donor can view all the blood requests on the donor dashboard. When a donor accepts a request, his/her acceptance is communicated from the React app to the backend via a PATCH request to the specific endpoint for that request. The server then validates the action and securely triggers a bidirectional exchange of contact information, revealing the seeker's phone number to the donor and vice-versa. The donor also gets the exact location of the seeker through the 'Navigation' option available on his dashboard, only upon this explicit confirmation.

5 - Navigation: Following the acceptance of a blood request by a donor, the React frontend dynamically renders an interactive map interface using the Google Maps JavaScript API, on the Donor dashboard. It automatically plots the route from the donor's current location to the designated hospital/seeker's location. This helps in donor's easy navigation and facilitates timely arrival.

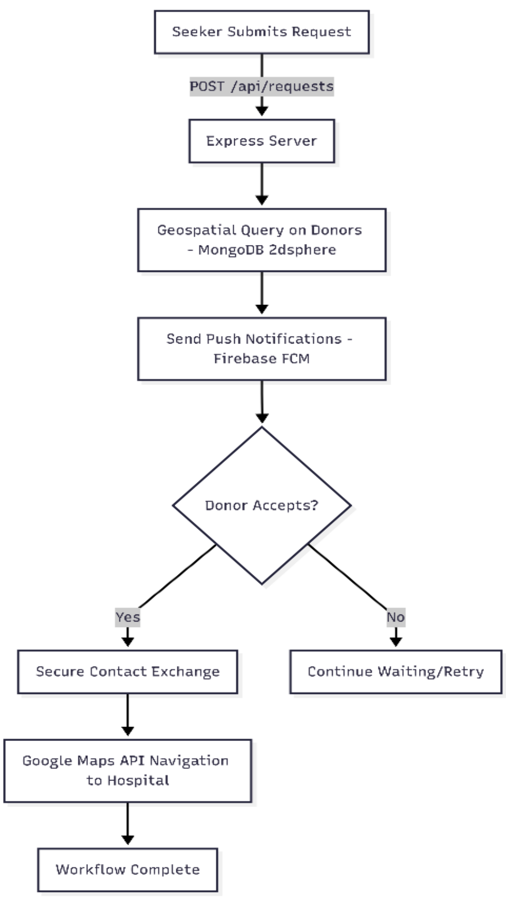

Fig. 4. Seeker Request and Acceptance

V. IMPLEMENTATION DETAILS

The whole platform has been implemented using the JavaScript ecosystem of the MERN stack. This allows for easy execution of the designed features and maintaining the efficiency of the system.

Frontend (React): The frontend has been built with React App for bootstrapping. The React Context API manages global state like user authentication. The frontend is designed as a Progressive Web App to offer app-like features such as offline functionality and home screen installation.

Backend (Node.js + Express): The backend is based on the use of Node.js and Express. The middleware employed by the server to provide security headers, cross-origin request to the frontend, express rate-limit to prevent API abuse, and elegant MongoDB object modelling.

Database (MongoDB Atlas): Database is deployed on MongoDB Atlas, the official cloud platform regarding MongoDB. This offers automatic backups, scaling and high availability without incurring the overhead of the manual database management.

Security: Hashing of passwords has been done using bcryptjs, which is of high standards. All API endpoints have authentication middleware. Secrets are stored in the environment variables and not the codebase.

VI. Results and Discussions

The practical implementation of the platform along with the performance results obtained during system testing are discussed below. The system was evaluated to ensure that it works efficiently and responds quickly enough to meet the urgent needs of emergency medical situations.

A. System Implementation and Visual Workflow

Rakt Setu has been successfully implemented using the MERN stack architecture. Various modules of the web app - from user registration to the blood request fulfillment are shown below:

User Onboarding: The onboarding interface (shown in Figure 5) allows users to register themselves on the Rakt Setu platform as donors and seekers by providing essential demographic data and blood group details. The onboarding process is aided with OTP based authentication.

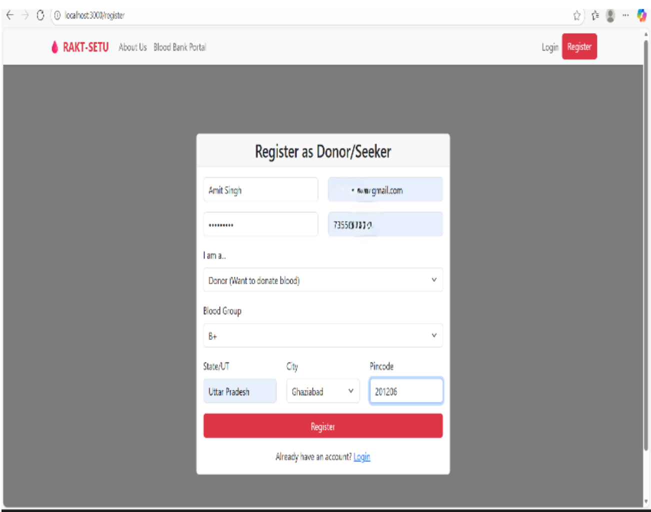

Fig. 5. User registration

Seeker Dashboard: Figure 6 shows the seeker dashboard which displays the real-time status of an active blood request along with a list of blood banks having the stock of matching blood.

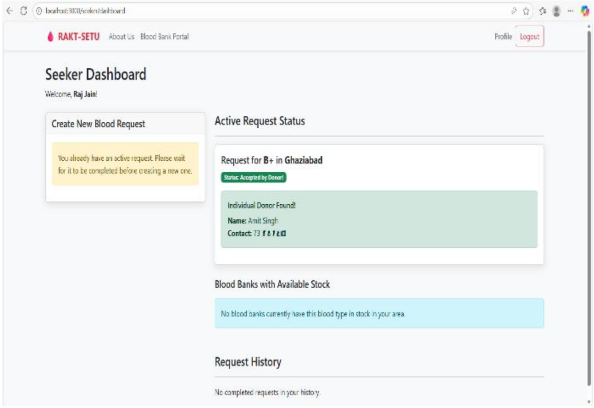

Fig. 6. Seeker Dashboard

Donor Dashboard: Figure 7 illustrates the donor dashboard, highlighting urgent local requests and providing an integrated navigation button for immediate response, once a request is accepted.

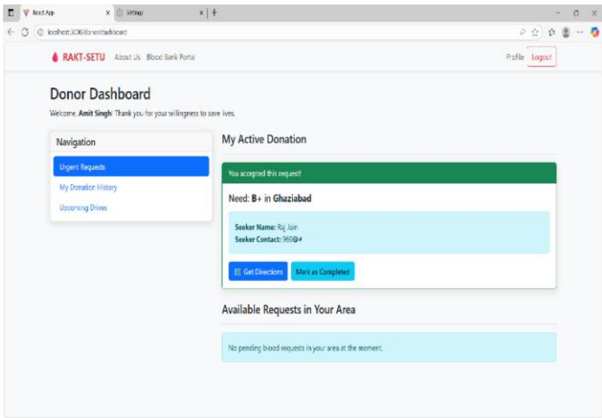

Fig. 7. Donor Dashboard

B. Empirical Performance Analysis

The system was tested for latency, delivery speed, and scalability to experimentally validate the research findings:

System Latency (Matching Speed):

We measured the time the Node.js backend takes to perform a geospatial query on the MongoDB database using the 2dsphere index. According to the testing results, the MongoDB 2dsphere index allows the backend to identify compatible donors within a 15km radius in less than 50ms.

Notification Delivery Time:

The speed of communication is critical for saving lives. We tracked the time from request submission to the donor receiving an alert via Firebase Cloud Messaging (FCM). Almost 95 out of every 100 alerts took less than 2 seconds to reach the donor devices and the 2nd email alerts took an average of 5 seconds.

Scalability and Reliability:

The system was able to manage a considerable amount of traffic because of Docker containerization. The application also maintains the same volume of memory regardless of where it is running. The Node.js server was able to handle a high number of 500 parallel geospatial queries, without a major decline in the response rate.

VII. Recommendations and Future Directions

This research is a solid basis of refining real-time blood donation and request management. It also reveals some strategies of improving the Rakt Setu platform to use it on a long-term basis. Next in line are the methods by which the participation of donors can be enhanced and the overall performance of the system can be enhanced.

A. Technical Improvements

1. Application of AI: Artificial Intelligence is applicable in finding donors who have higher chances to positively respond to a request depending on how they did respond in the past.

2. Application of Blockchain: Blood units can be tracked with the help of a digital ledger. This would contribute towards making supply chain transparent and safe.

3. Developing Native Apps: Android and iOS apps will be created to help in utilizing superior location tracking and push notifications. This would enhance the experience of the users.

B. Strategic Growth

1. Integration with Government Portals: The system can be directly connected to portals such as E-Rakt Kosh [5] to get more information about the inventories.

2. Gamification: It is possible to introduce a reward system that will motivate individuals to donate more.

VIII. Conclusion

Rakt Setu is a complete web-based application to support blood donation and management of requests. It deals with other pressing issues of blood seekers, when there is an emergency situation and an urgent need of blood. This piece of research has enhanced the existing technology in a much better way and simplified the process of donating as well as requesting blood online. MERN stack architecture offers good performance, scalability, and is an architecture that is not difficult to maintain. Containerization with Docker provides an opportunity to be deployed consistently across the board. One of the key findings in this study is the use of the 2dsphere indexing of the MongoDB to match donors based on their proximity. This is what allows real-time connections and this reduces the emergency response time by matching donors who are compatible within milliseconds. Secure two-way communication protocol is used to maintain the privacy of the user and to make sure that the contact information is shared upon a blood request being accepted by a donor. The web application is also linked with Google Maps API that would allow smooth navigation to confirmed healthcare facilities and secure donor safety and reliability of the procedure. The partnerships with healthcare institutions and NGOs are some of the strategic partnerships that make blood donation activities reliable. Rakt Setu is an enormous enhancement of the old platforms. In comparison with the old platforms that operated in isolation and executed few functions, Rakt Setu develops a framework that coordinates the entire operation including urgent request to successful donation. It unites the seekers, donors and the blood banks at one platform and all are interconnected where they can communicate directly without risk of privacy. This approach may alter the response to blood emergencies in communities particularly those that handle severe blood shortages.

References

[1] Apollo Hospitals, "Donate Blood Safe Life", February 2025. [Online]. Available: https://www.apollohospitals.com/procedures/donate-blood-save-life

[2] World Health Organization (WHO), "Blood Safety and Availability," Fact Sheet, June 2022. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/blood-safety-and-availability

[3] Friends2Support, "I am for Blood Donation," Friends2Support International Voluntary Blood Donors Organization. [Online]. Available: https://www.friends2support.org/

[4] The American Red Cross, "Blood Donor App." [Online]. Available: https://www.redcrossblood.org/donate-blood/blood-donor-app.html

[5] Ministry of Health and Family Welfare, Government of India, "E-Rakt Kosh: National Blood Donation Service." [Online]. Available: https://eraktkosh.in/

[6] Sachdev S, Mishra SK, Marwaha N, Avasthi A., Study of knowledge and attitude among college-going students toward voluntary blood donation from north India. J Blood Med. 2016;(7):19–26 .

[7] A. F. Osorio, S. C. Brailsford, and H. K. Smith, "A Structured Review of Quantitative Models of the Blood Supply Chain," Computers & Industrial Engineering, vol. 101, pp. 383-397, Nov. 2016.

[8] Prof. Gouri Shukla, Sudarshan Gosavi, , Vishal Chaugule, Rohit Dhavare and Soham Galfade, " BloodLink Navigator System" in International Journal of Research Publication and Reviews, Vol (6), Issue (3), March (2025).

[9] Sangeetha, M., Dhivya, A., Ramya, R., & Sharmila N., Blood donor searching android application. International Journal of Health Sciences, 6(S1), 13492–13503, (2021).

[10] Gupta, S., & Verma, K., Enhancing blood donor engagement using mobile-based applications: A case study. Health Information Systems Research, 9(1), 88-102, (2020).

[11] Kumar, R., & Sharma, P., Advancements in digital blood bank management: A review of emerging technologies. International Journal of Healthcare Technology, 12(4), 45-59, (2021).

[12] Ashita Jain, Amit Nirmal, Nitish Sapre, Prof Shubhada Mone, Online Blood Bank Management System Using Android. International Journal of Innovative Studies in Sciences and Engineering Technology(IJISSET),ISSN 2455- 4863, (2016).